Experiment:-3

Problem Statement:-

Develop a network simulator using Cisco Packet Tracer to analyze packet delay, loss, and end-to-end throughput. Implement various routing algorithms and measure their impact on network performance under different traffic conditions.

Aim:- **Network Simulator for Analyzing Packet Delay, Loss, and Throughput**

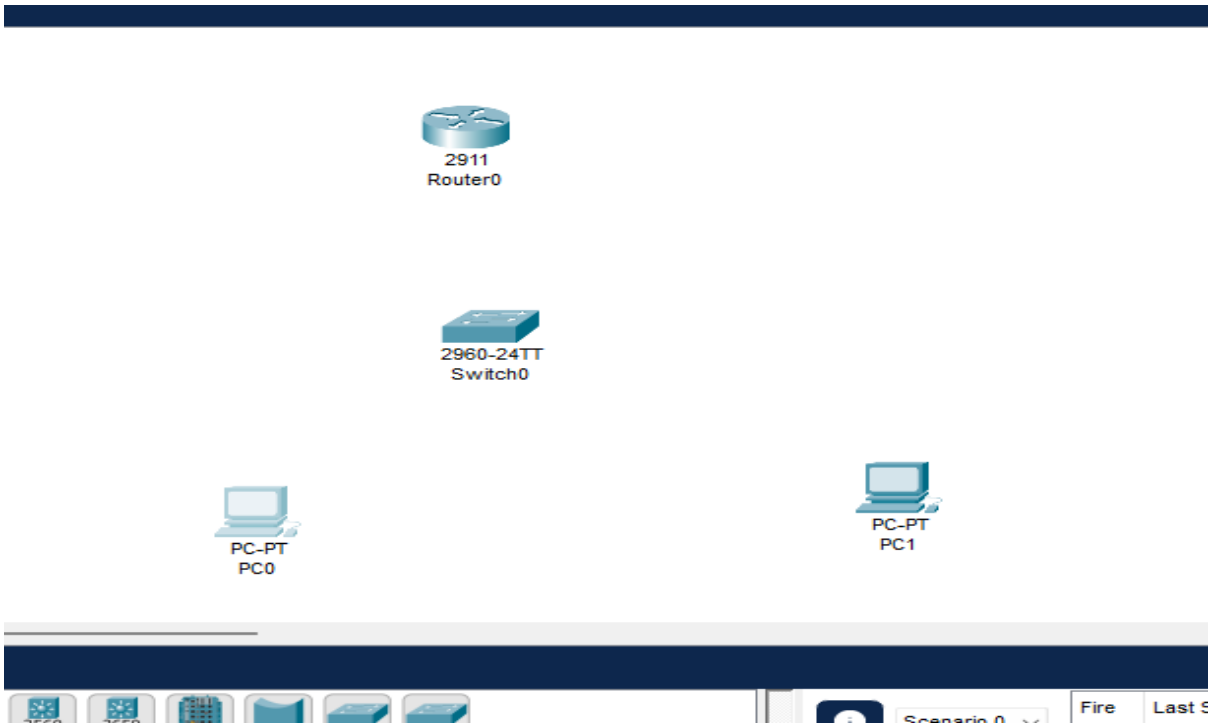
Procedure:-

Step 1: Open Packet Tracer

- Click **File → New**

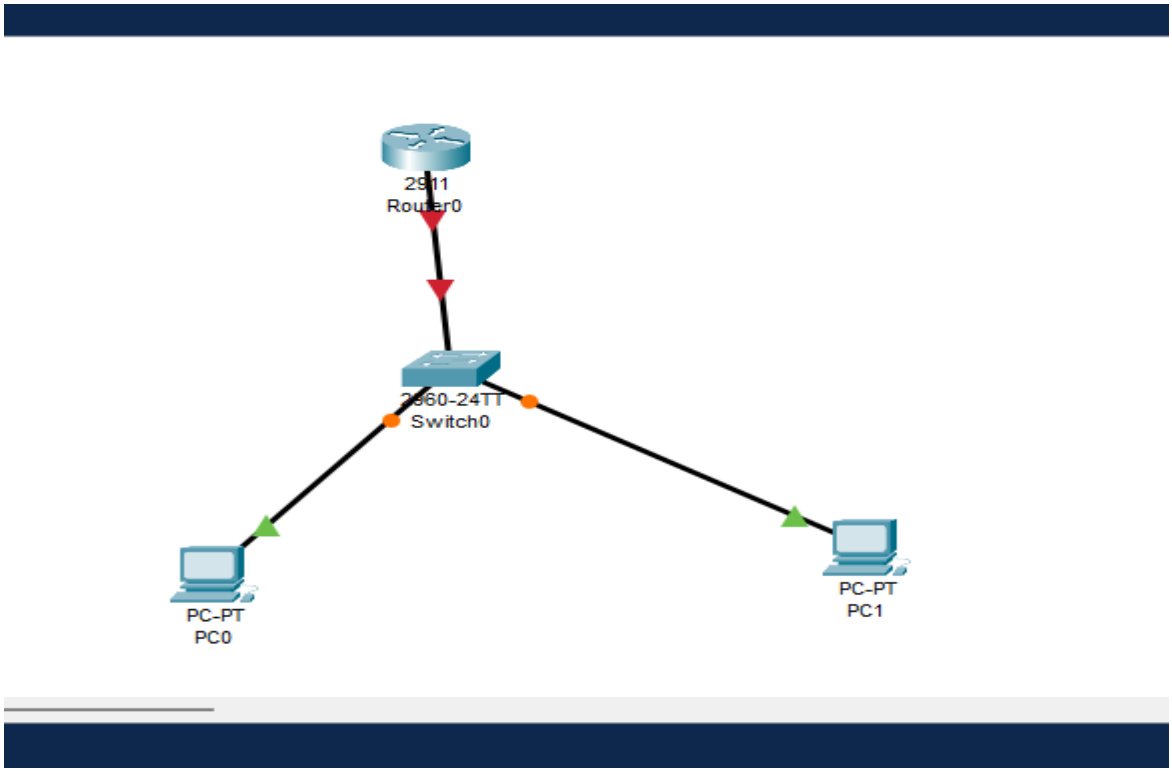
Step 2: Add Devices to Workspace

Drag and drop devices:-

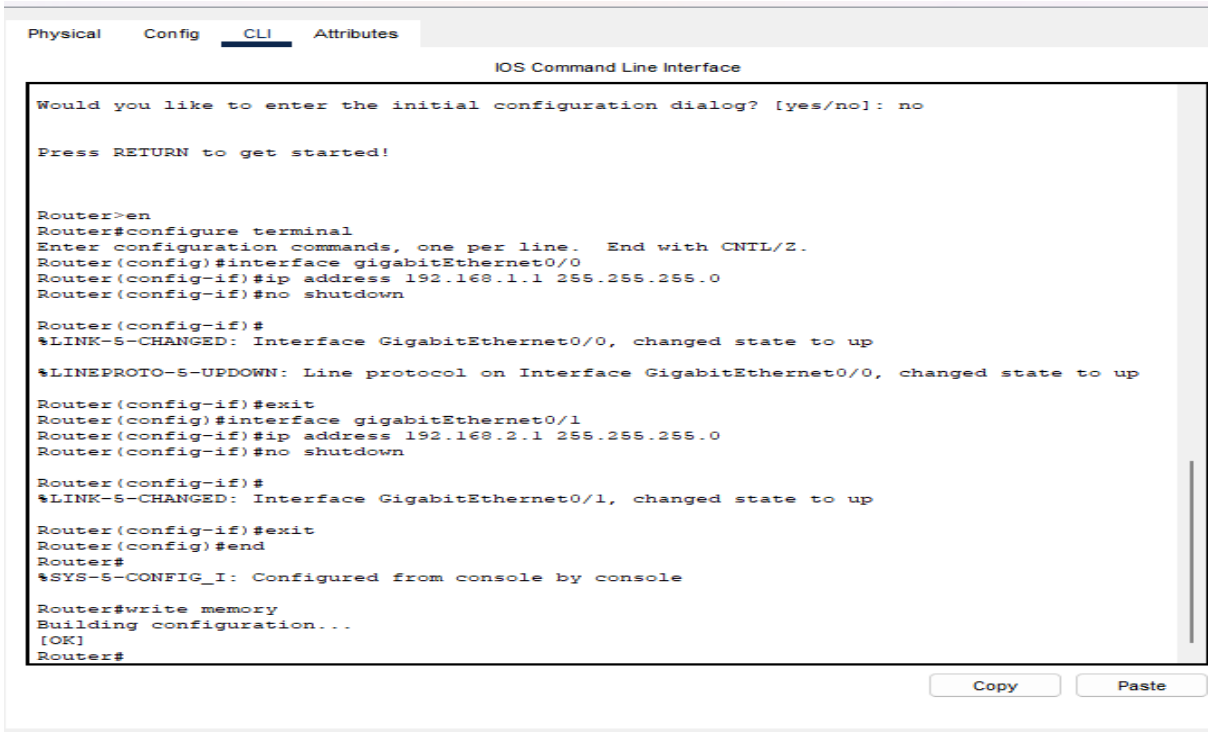


Step 3: Connect Devices with Cables

Use **Copper Straight-Through** cable:



Step 4: Click Router → CLI tab



Step 5: Verify Router

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
up


Router>
Router>en
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Step 6: Click each PC → Desktop → IP Configuration

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 192.168.10.2

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

Automatic Static

IPv6 Address /

Link Local Address FE80::201:C7FF:FE25:D006

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication MD5

Username

Password

Step 7: Add Static Routes on Router

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 gigabitEthernet0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
Router(config)#ip route 192.168.2.0 255.255.255.0 gigabitEthernet0/1
%Default route without gateway, if not a point-to-point interface, may impact
performance
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#
```

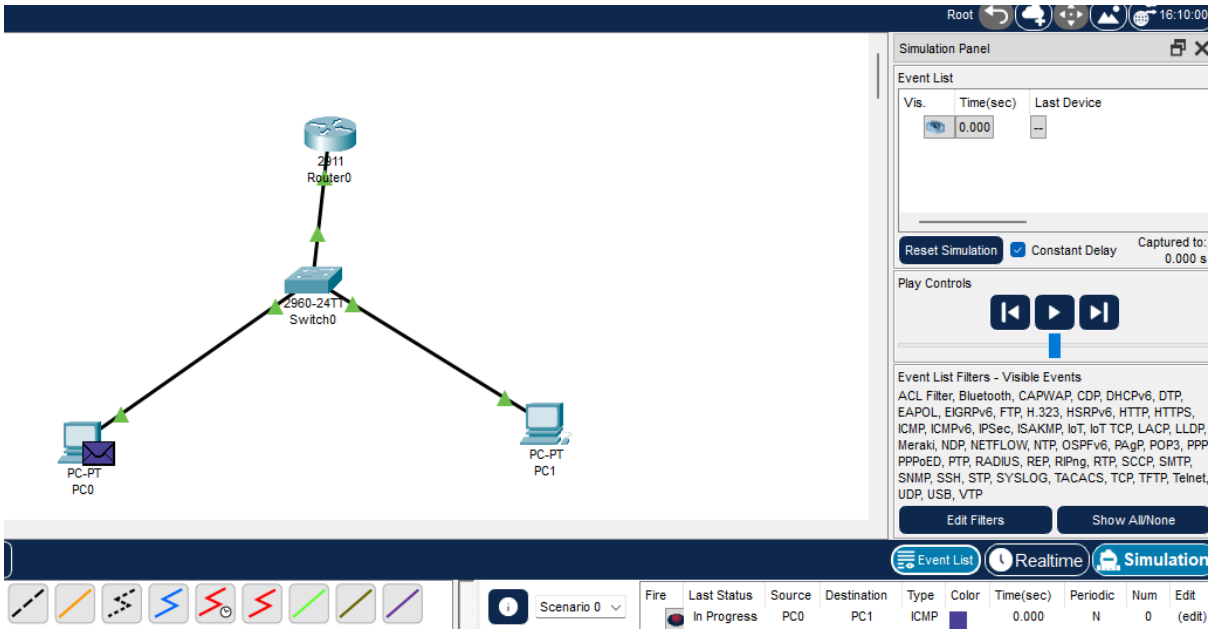
Step 8: Ping from PC0

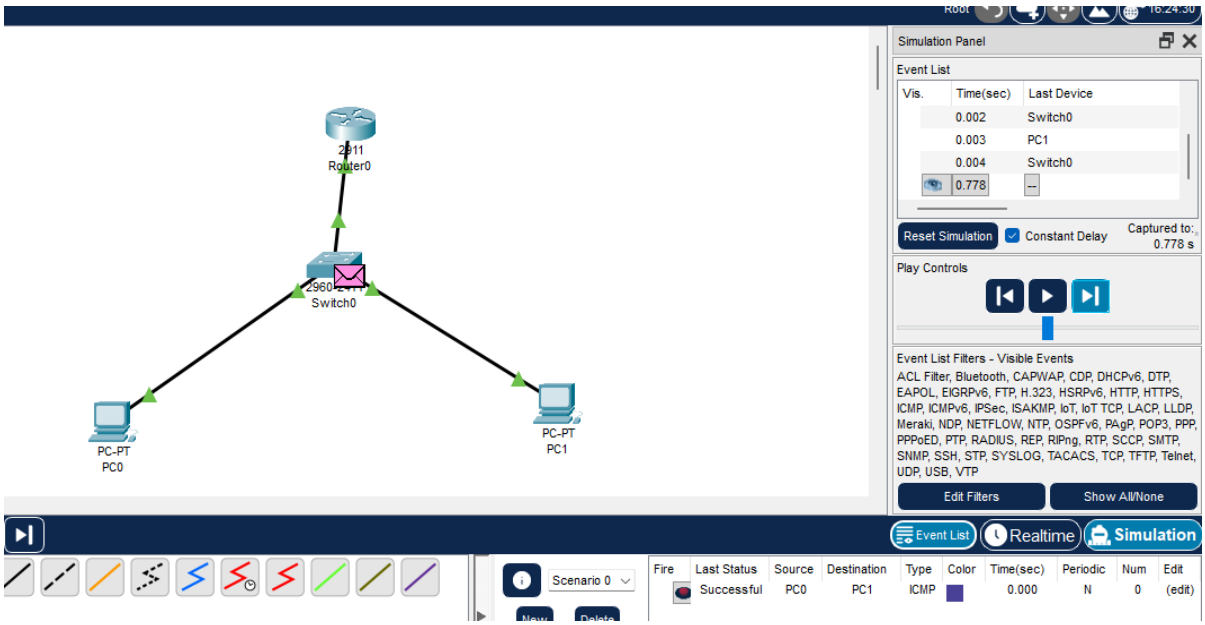
- Click **PC0 → Desktop → Command Prompt**

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Step 9: Use Add Simple PDU

- Click the **envelope icon** (Add Simple PDU) in the toolbar
- Click **PC0** then click **PC1**
- A packet will appear on the workspace



Step 10: Switch to Simulation Mode

- Click **Simulation** tab (bottom right)
- Press **Play** or **Step** button
- Watch packets travel PC0 → Switch0 → Router → Switch1 → PC1